Chapter 13
Feedback and Control

Abstract The fundamentals of feedback and control systems are presented, along with how they operate in healthy and nonhealthy people. Specific examples include the regulation and control of body temperature and blood pressure, including during exercise, and of pupil size.

We consciously control our thought processes to achieve physical goals, in part by using our senses to provide response for feedback to control our actions. For example, when we place an object on a table we control our actions by using our eyes to provide feedback on the relative positions of the object and the table as it approaches the table and our sense of touch to provide feedback as we place it on the table. Such feedback and control is also important in all manufacturing processes. Our bodies function amazingly well because of constant feedback and control processes.

One very essential feedback and control system is *homeostasis*, which technically is the stability of the chemical and physical conditions of the fluid surrounding the body cells. This extracellular fluid constitutes about one third of the total body fluid, and includes the blood plasma and interstitial fluid—the tissue fluid that is in the spaces between the cells. The other two thirds of the fluid is intracellular. The extracellular fluid is controlled to regulate (a) body temperature, (b) pressure in blood vessels, (c) oxygen and carbon dioxide concentrations, (d) pH, (e) the concentrations of ions, such as Na^+, K^+, and Ca^{2+}, (f) volume, (g) osmolality (water/dissolved particle ratio), and (h) the organic nutrient concentrations, such as glucose.

We have many other important control systems. Our eye irises open and close in response to light levels. The ciliary muscles control the focal length of the crystalline lenses in our eyes to focus on objects (accommodation, Fig. 11.22). The force we use to grasp objects depends on the normal force feedback we get from it (and if the object "gives" we apply less force). During exercise the intercostal and abdominal muscles that control breathing are regulated by the brain, which receives input from chemical receptors in the blood, mechanical and metabolic receptors in skeletal muscles, as well as from receptors in the lungs (Fig. 13.1). Similarly, during exercise the brain receives input from the heart and muscles and then sends signals to control the heart

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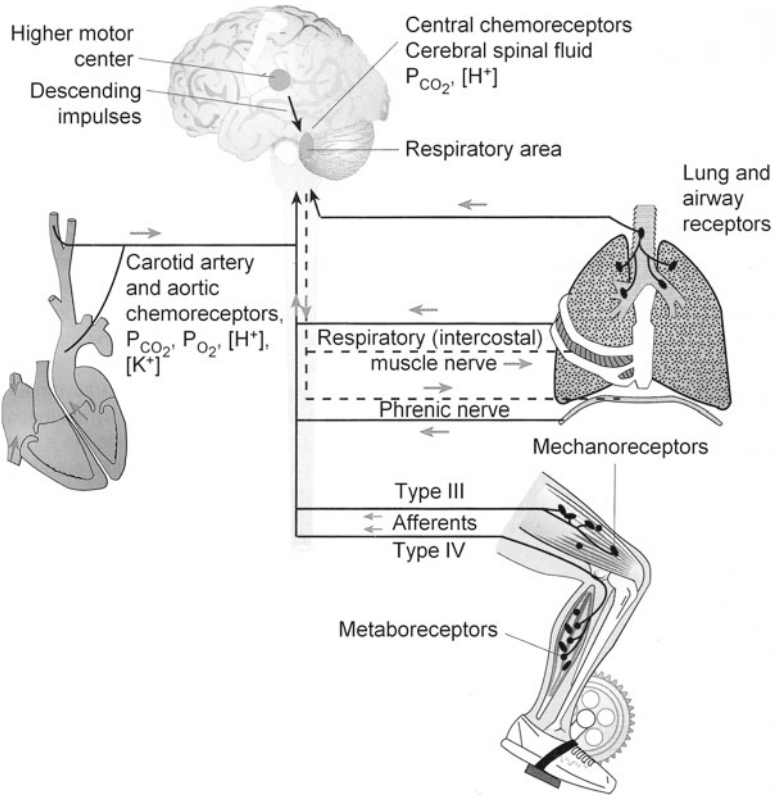


Fig. 13.1 Control of the respiratory (ventilation) system during exercise. The respiratory area in the brain gets signals from the brain motor region, central chemoreceptors, and other receptors, including carotid artery and aortic chemoreceptors, lung intercostal muscle and diaphragm receptors, and skeletal muscle mechanical and metabolic receptors. During exercise this results in increased neural activity to the intercostal muscles and diaphragm through the phrenic and intercostal nerves. This increases the rate and depth of breathing, which regulates arterial oxygen, carbon dioxide, and pH (From [4]. Used with permission)

and blood vessels to increase cardiac output and blood pressure (Fig. 13.6 below). Body heat balance and temperature are also controlled during exercise (Fig. 13.4 below). Of course, these control systems are also essential during resting. For general discussions about such control systems and control in the body see [1, 4, 6, 7, 10, 12, 15, 19, 21].

13.1 Basics of Feedback and Control

Figure 13.2 is a block diagram depicting *feedback and control*. There is a sensor or receptor that measures a quantity (stimulus), such as blood pressure by baroreceptors. The signal from this receptor is transmitted to an integration center, such as the

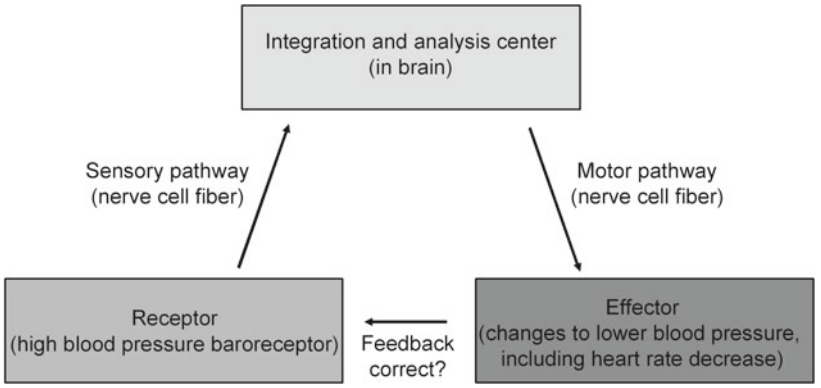


Fig. 13.2 Feedback and control in the body, with an example of controlling high blood pressure in parenthesis (Based on [19])

transmission of signals from sensory centers to the brain by nerves. The integration center induces an effect. This response is transmitted by nerves (motor pathways) to an effector control center. There is some effect or response of interest, such as changing the heart rate or stroke volume to change the blood pressure. The success of this control is determined by measuring the response, the new blood pressure, which provides feedback for the control.

Negative feedback reverses the direction of the change of a variation, to keep the measured parameter near the desired *set point*. This type of feedback is very common in the body. There is usually an operating range centered about this set point bounded by allowable values. Figure 13.3a shows that the effector is activated to correct the parameter when it wanders above the highest value allowed or below the lowest value allowed. Room thermostats usually operate in the same manner, with a several degree operating range about the set point; our bodies also have thermostats.

Positive feedback causes the effector to produce more of a change in the same direction that the parameter is already changing, as in Fig. 13.3b. This type of feedback is rare in the body. One example is suckling which leads to the production of more milk in mothers. Another occurs during the final stages of death by chronic congestive heart failure [16]. With this condition, the cardiac muscle is not able to contract enough to attain the needed stroke volume. Fluid backs up in the veins,

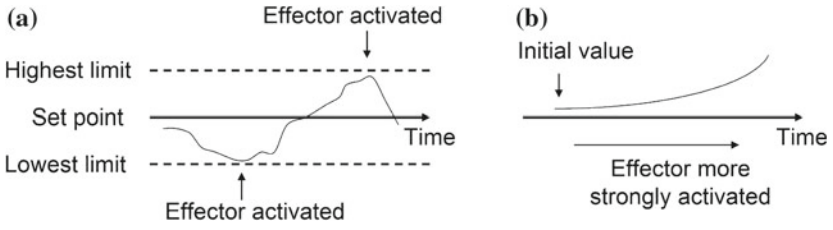


Fig. 13.3 **a** Negative and **b** positive feedback (Based on [19])

causing increased pressure on the lungs and other organs and not enough blood is circulated to the organs. Instead of returning the system to a better set point, positive feedback worsens heart performance and, as damaging, causes the blood volume to increase steadily much beyond the optimal value. Poor heart performance leads to the thickening of the cardiac muscle, but the new muscle is of low quality and decreases heart performance even further. The already limited blood flow to the cardiac muscle becomes increasingly insufficient to keep the increased amount of muscle healthy, which deteriorates performance even more. Concurrently, blood volume increases, increasing the load on the damaged heart, because of the response of the kidneys. The kidneys filter out some salt and water and normally remove them from circulation. For a healthy person this maintains the overall blood volume. For someone with congestive heart failure, such removal would decrease blood volume a bit and lessen the load on the heart, which would be good. However, the kidneys sense the lower blood flow and instead return to circulation the salt and water it has filtered out in an effort to increase the low blood volume to normal levels. This undesirably increases the blood volume, which increases the load on the heart, which causes the heart to deteriorate further and pump less blood, which causes the kidneys to increase total blood volume, and so on, accelerating the death process.

13.1.1 Control Theory (Advanced Topic)

Feedback is needed to control a parameter P, such as blood pressure or temperature, to keep it near a set point P_0. The level of feedback is often proportional to the difference between the current parameter value and the desired set point [15]. The gain g is the proportionality factor and this determines the magnitude of the response. This response causes a change $\mathrm{d}P$ in time $\mathrm{d}t$ and so

$$\mathrm{d}P = -g(P - P_0)\mathrm{d}t. \tag{13.1}$$

This feedback is not instantaneous. It is delayed by a time τ_d due to delays associated with sensing, neuronal transmission to the brain, processing in the brain, neuronal transmission to the effectors, and how fast the effectors can cause a change. Consequently, a change at time t is due to feedback provided from sensors at time $t - \tau_\mathrm{d}$, and so (13.1) changes to

$$\mathrm{d}P(t) = -g(P(t - \tau_\mathrm{d}) - P_0)\mathrm{d}t, \tag{13.2}$$

or defining $p = P - P_0$

$$\frac{\mathrm{d}p(t)}{\mathrm{d}t} = -gp(t - \tau_\mathrm{d}). \tag{13.3}$$

The desired response is a monotonic decrease of the magnitude of p, a stable oscillation of p about 0 (and therefore of P about P_0), or some combination of the

two. *What values of g and τ_d can produce this desired stable negative feedback?* Obviously g must be positive and it may seem that the larger the gain the better, but we will see that large values of the gain are not always desirable.

Let us try to solve (13.3) by substituting the possible solution $p(t) = p\ \exp(zt)$ into it. In general, z can be a complex number, with $z = x + \mathrm{i}y$. The imaginary term y gives oscillatory behavior about the set point, while $x < 0$ gives exponential decay to the set point and $x > 0$ gives an exponential increase away from the set point, and this last case is undesirable. Equation (13.3) then becomes

$$z = -g\ \exp(-z\tau_d). \tag{13.4}$$

Reasonable solutions can exist for $x \leq 0$. Because $\exp(\mathrm{i}q) = \cos q + \mathrm{i}\sin q$, (13.4) becomes separate equations for the real and imaginary parts.

$$x = -g\ \exp(-x\tau_d)\cos y\tau_d, \tag{13.5}$$

$$y = g\ \exp(-x\tau_d)\sin y\tau_d. \tag{13.6}$$

These are transcendental equations that in general must be solved numerically. We will, however, be able to solve them analytically in special regimes.

The solution with $x = 0$ is one that oscillates about the set point. Using $x = 0$ in (13.5) leads to $0 = -g\cos y\tau_d$, which gives $y = \pi/2\tau_d$. Using $x = 0$ in (13.6) leads to $y = g\sin y\tau_d$ and with $y = \pi/2\tau_d$, this becomes $\pi/2\tau_d = g\sin((\pi/2\tau_d)\tau_d) = g$. This means

$$g\tau_d = \frac{\pi}{2}, \tag{13.7}$$

and so if the delay time is τ_d, there is an oscillatory response when the gain $g = \pi/2\tau_d$. This is an acceptable solution.

Now let us say that x has a small magnitude but is not necessarily 0. Equation (13.5) still gives $y \simeq \pi/2\tau_d$ and so $\sin y\tau_d \simeq 1$. Because $\exp(q) \approx 1 + q$ for $|\,q\,| \ll 1$, (13.6) becomes $y \simeq g(1 - x\tau_d)\sin y\tau_d \simeq g(1 - x\tau_d)$ and so

$$x \simeq \frac{1}{\tau_d}\left(1 - \frac{\pi}{2g\tau_d}\right). \tag{13.8}$$

For $g\tau_d < \pi/2$, we see that $x < 0$ and the parameter P approaches the set point as an oscillation with a magnitude that decreases with time. For $g\tau_d = \pi/2$, it is clear that $x = 0$ and the parameter P oscillates about the set point, as shown above. For $g\tau_d > \pi/2$, we see that $x > 0$ and the parameter P oscillates about the set point with a magnitude that increases with time and this is unstable. Therefore, this stability criterion sets an upper limit to the possible gain for a given delay time, given by

$$g\tau_d \leq \frac{\pi}{2}. \tag{13.9}$$

13.2 Regulation of the Body

13.2.1 Regulation of Temperature

The regulation of body temperature is closely tied to the discussion in Chap. 6 of the production of heat by the metabolism and the modes of heat loss from the body. The normal core body temperature is 37 ± 2 °C (98.6 ± 3.6 °F). Large increases and decreases mean trouble. Far above normal body temperatures, at 41 °C (106 °F) the central nervous system begins to deteriorate and convulsions occur. At 45 °C (115 °F), proteins denature, followed by death. For lower than normal temperatures, at 33 °C (91 °F), nervous functions are depressed to the point that consciousness is lost. At 30 °C (86 °F), the temperature regulation system fails. At 28 °C (82 °F), there is cardiac fibrillation, leading to death. The body is quite good at regulating core temperature, as is clear from Fig. 6.20, but it has its limits and can fail, as is clear from Figs. 6.21, 6.22 and 6.23. Figure 13.4 illustrates how the receptors in the skin and in the core sense temperature and send information to the brain. Temperatures

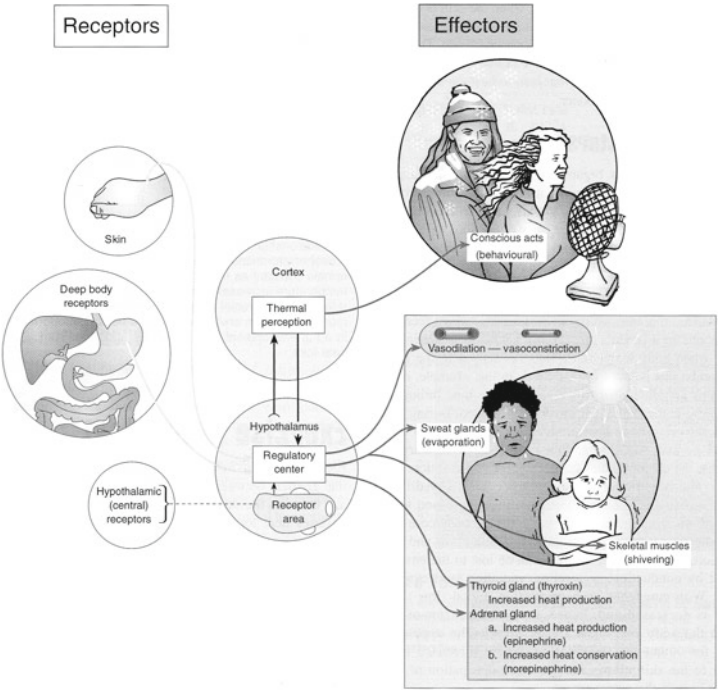


Fig. 13.4 The thermoregulatory system, including input to the brain from receptors and output from the brain to effectors. The hypothalamic center relays information to the effector organs to stimulate the heat production and conservation processes shown when the core temperature receptors differ from the 37 °C set point (From [4]. Used with permission)

outside the range of normality lead to conscious acts and automatic activity by the body to correct the temperatures.

Each person has a set point temperature, which varies by $\simeq \pm 0.3\,^{\circ}\mathrm{C}$ ($\pm 0.5\,^{\circ}\mathrm{F}$) among people [3]. The mean set point for skin temperature is approximately 33 °C (91.4 °F). (In Chap. 6 we used a skin temperature of 34 °C.) Skin temperatures above 34.5 °C (94 °F) cause active sweating, while those below 30 °C (86 °F) lead to the increased metabolic activity associated with shivering.

Recalling our discussion in Chap. 6, the conservation of energy implies a conservation of heat creation and loss

$$\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{net}} = \left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{metabolism}} + \left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{passive loss}} + \left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{body controlled}}. \tag{13.10}$$

The left side is the net heating rate due to all modes of heat production and heat loss. The first term on the right-hand side is the heat production from metabolic activity and physical activity; it is >0. The second term is heat loss due to passive modes, such as from the skin, radiation, convection, etc., it is <0. The third term is that due to body-controlled heating (>0) and/or heat removal (<0) [1].

Our discussion in Chap. 6, as in (6.5) and (6.36), showed us how body temperature changes with a net flow of heat into or out of the body, according to

$$\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{net}} = C\frac{\mathrm{d}T}{\mathrm{d}t} = (60\ \text{kcal}/^{\circ}\text{C})\ \frac{\mathrm{d}T}{\mathrm{d}t}, \tag{13.11}$$

where C is the heat capacity $= 60\,\text{kcal}/^{\circ}\text{C}$ for a 70 kg person. The heat capacity $C =$ mass (=70 kg) $\times$ specific heat c (= 0.83 kcal/°C-kg). We know that the minimum heating rate due to the metabolism is the BMR, so it can be written as

$$\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{metabolism}} = f(\text{BMR}), \tag{13.12}$$

where f is the activity factor—a multiplier between 1 and about 20, and BMR = 70 kcal/h.

If there is no loss of heat (and the last two terms of (13.10) are zero), we see that

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = f(\text{BMR}) = f\ 70\ \text{kcal/h} \tag{13.13}$$

so

$$\frac{\mathrm{d}T}{\mathrm{d}t} = f\frac{\text{BMR}}{C} = f\frac{70\ \text{kcal/h}}{60\ \text{kcal}/^{\circ}\text{C}} = 1.2f\ ^{\circ}\text{C/h}. \tag{13.14}$$

This is similar to (6.36). Without adequate means of heat loss we would not survive heavy exercise for long. With $f = 20$ for heavy exercise, we see that $\mathrm{d}T/\mathrm{d}t = 24\,^{\circ}\mathrm{C/h}$.

Now let us consider what happens when there is passive loss of heat only, due to radiation and convection, with no control. From (13.10), the loss of heat due to radiation and convection is

$$\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{passive loss}} = -(Ah_\mathrm{r} + Ah_\mathrm{c})(T_\mathrm{skin} - T_\mathrm{room}) = -\lambda(T_\mathrm{skin} - T_\mathrm{room}), \tag{13.15}$$

where $\lambda = A(h_\mathrm{r} + h_\mathrm{c})$. For a person with a body area of $1.5\,\mathrm{m}^2$, the radiation parameter $Ah_\mathrm{r} \sim 12\,\mathrm{kcal/h\text{-}^{\circ}C}$. The convection parameter is $Ah_\mathrm{c} \sim 13\,\mathrm{kcal/h\text{-}^{\circ}C}$ for a nude person in the presence of a 9 m/s wind speed. This is much higher than typical values for clothed people. For a nude person in a 15 m/s breeze it is larger by a factor of 5 than the value used here.

Equation (13.15) now becomes

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = f(\mathrm{BMR}) - \lambda(T_\mathrm{body} - T_\mathrm{room}) \tag{13.16}$$

or

$$(60\ \mathrm{kcal/^{\circ}C})\frac{\mathrm{d}T}{\mathrm{d}t} = f(70\,\mathrm{kcal/h}) - (25\ \mathrm{kcal/h\text{-}^{\circ}C})(T_\mathrm{body} - T_\mathrm{room}), \tag{13.17}$$

where we have called the body temperature T_body and are now ignoring the differences between the skin and core temperatures. These equations can be solved by substituting the form

$$T_\mathrm{body}(t) = \alpha + \beta\exp(-t/\tau_\mathrm{r}), \tag{13.18}$$

where τ_r is the characteristic response time. This substitution gives

$$C\beta(-1/\tau_\mathrm{r})\exp(-t/\tau_\mathrm{r}) = f(\mathrm{BMR}) - \lambda(\alpha + \beta\exp(-t/\tau_\mathrm{r}) - T_\mathrm{room}) \tag{13.19}$$

$$= f(\mathrm{BMR}) - \lambda(\alpha - T_\mathrm{room}) - \lambda\beta\exp(-t/\tau_\mathrm{r}). \tag{13.20}$$

Because this must be valid for all time t, we can equate the coefficients of the $\exp(-t/\tau_\mathrm{r})$ terms to get

$$C\beta(-1/\tau_\mathrm{r}) = -\lambda\beta, \tag{13.21}$$

$$\tau_\mathrm{r} = \frac{C}{\lambda} = \frac{60\ \mathrm{kcal/^{\circ}C}}{25\ \mathrm{kcal/h\text{-}^{\circ}C}} = 2.4\ \mathrm{h}, \tag{13.22}$$

and then equate the constant terms to get

$$f(\text{BMR}) = \lambda(\alpha - T_{\text{room}}), \tag{13.23}$$

or

$$\alpha = T_{\text{room}} + f\frac{\text{BMR}}{\lambda} = T_{\text{room}} + f\frac{70\ \text{kcal/h}}{25\ \text{kcal/h-°C}} = T_{\text{room}} + 2.8f\ ^\circ\text{C}. \tag{13.24}$$

At $t = 0$ we will call $T_{\text{body}}(t = 0) = T_{\text{body,initial}}$ (which is the initial condition), so,

$$T_{\text{body}}(0) = T_{\text{body,initial}} = \alpha + \beta \tag{13.25}$$

and therefore

$$\beta = T_{\text{body,initial}} - \alpha = T_{\text{body,initial}} - T_{\text{room}} - 2.8f\ ^\circ\text{C}. \tag{13.26}$$

The final solution is

$$T_{\text{body}}(t) = T_{\text{room}} + 2.8f\ ^\circ\text{C} + (T_{\text{body,initial}} - T_{\text{room}} - 2.8f\ ^\circ\text{C})\exp(-t/\tau_{\text{r}}). \tag{13.27}$$

This is an example of *open loop* (*or open cycle*) heating, because there is no feedback and control. The problems in this temperature regulation system are clear from (13.27). The final steady-state temperature (i.e., that for $t \gg \tau_{\text{r}} = 2.4\,\text{h}$) is

$$T_{\text{body,steady state}} = T_{\text{room}} + f\frac{\text{BMR}}{\lambda} = T_{\text{room}} + 2.8f\ ^\circ\text{C}. \tag{13.28}$$

If $T_{\text{room}} = 21\,^\circ\text{C}$ (69.8°F $\simeq 70\,^\circ$F) and we were resting, so $f \sim 1$, our steady-state body temperature would be $\approx$24 °C (75 °F), which is much lower than our desired body temperature. If we were exercising pretty hard, so $f = 10$, our steady-state body temperature would be $\approx$49 °C (120 °F), which is too high. We could consciously change λ a bit (through h_{c}: by decreasing it by putting on heavy clothes in the first case or by increasing it by increasing the wind speed on our nude bodies in the second case), and this would help a little. Ideally, we would want to fix

$$T_{\text{body,steady state}} - T_{\text{room}} = 37\,^\circ\text{C}\ (98.6\,^\circ\text{F}) - 21\,^\circ\text{C}\ (69.8\,^\circ\text{F}) = 16\,^\circ\text{C}\ (28.8^\circ\text{F}) \tag{13.29}$$

$$= f\frac{\text{BMR}}{\lambda} = 2.8f\ ^\circ\text{C}, \tag{13.30}$$

so, $f \sim 6$ with our current λ. We could regulate temperature by changing f or λ. This means to maintain our body temperature we would have to constantly engage

in moderate activities, such as slow swimming (for the same λ). If we bundled up some (decreased λ), we could lower f a bit, to say $f = 4$, which would mean we would have to still engage in activities, such as moderately fast walking—even while we were asleep (such as constant sleep walking).

Another negative feature of the consequences of (13.27) is that it has a time constant $\tau_r = 2.4\,\text{h}$ that is much too long. If we decided to change our activity level (and therefore change f) it would take about 5 h for our body temperature to stabilize to the new steady-state temperature; this is unreasonable. These seemingly ridiculous results cannot be attributed to the assumptions we have made in the heat gain and loss terms so far. We have neglected the last term in (13.10) which describes the feedback and control for *closed loop (or closed cycle)* temperature regulation. We need this to maintain body temperature within the normal range.

The last term in (13.10) $(\text{d}Q/\text{d}t)_{\text{body controlled}}$ describes the involuntary responses by the body to maintain a normal body temperature, which we can call the set point—the targeted temperature. When an increase in temperature is needed, this term is positive due to increased metabolic activity and motor activity, such as shivering (up to the summit metabolic rate, Chap. 6); this heat goes directly to the core and muscles. When a decrease in temperature is needed this control term is negative, e.g., due to sweating. The body can lose 1 L/h water by sweating; if all of this sweat evaporates (and is not wiped off), then with the 540 kcal/L heat of evaporation of water, the heat loss through sweating can be 540 kcal/h. Through vasomotor control, this control term can be positive or negative. If blood is directed to vessels closer to the skin, cooling is more effective and the last term in (13.10) is negative. If blood is diverted to vessels farther from the skin and closer to the core, cooling is less effective and the control term is positive. One reason cooling is less effective is the countercurrent heat exchange mechanism described in Fig. 6.19.

Figure 13.5 shows the rate of heat loss due to sweating as a function of the body core temperature. This loss is zero when the core is cooler than normal and increases fairly linearly with temperature as $K_{\text{sweat}}(T_{\text{body}} - 36.85\,^\circ\text{C})$ where $K_{\text{sweat}} \sim 750\,\text{kcal/h-}^\circ\text{C}$, when the core is higher than normal (for skin temperature

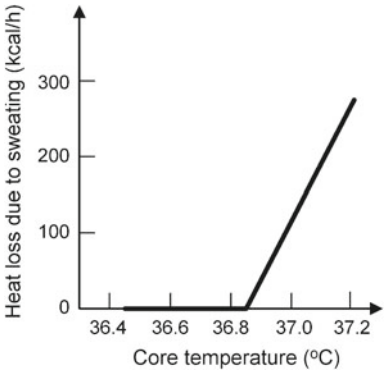


Fig. 13.5 Rate of heat loss due to sweating, for skin temperatures between 33 and 38 °C (Based on [1, 2])

between 33 and 38 °C). Now (13.10) becomes

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = f P_0 - \lambda(T_{\text{body}} - T_{\text{room}}) - K_{\text{sweat}}(T_{\text{body}} - 36.85\,^{\circ}\text{C}), \tag{13.31}$$

or

$$\begin{aligned}(60\ \text{kcal}/^{\circ}\text{C})\frac{\mathrm{d}T}{\mathrm{d}t} = {} & f(70\ \text{kcal/h}) - (25\ \text{kcal/h-}^{\circ}\text{C})(T_{\text{body}} - T_{\text{room}}) \\ & -(750\ \text{kcal/h-}^{\circ}\text{C})(T_{\text{body}} - 36.85\,^{\circ}\text{C}).\end{aligned} \tag{13.32}$$

Note that T_{body} in the passive heat loss term really refers to the skin temperature, while that in the control term is the core temperature. The solution to this equation has a form similar to (13.27) with the response time

$$\tau_{\text{r}} = \frac{C}{\lambda + K}. \tag{13.33}$$

Because $K/\lambda = 30$, the time constant is now smaller by a factor of 31, giving $2.4\,\text{h}/31 \simeq 4.7\,\text{min}$, which is much more reasonable.

Note that we have been able to see how the body can increase the gain to produce a response time that is reasonable on a physiological timescale. We have ignored the issue of the delay time τ_{d}, as discussed in the previous section. For this process at least $\tau_{\text{d}} \ll \tau_{\text{r}}$ and so the delay time is not important here.

Our bodies usually have a set point for the core temperature of about 37 °C. Sometimes when we are ill, bacterial toxins called pyrogens increase this set point. When this set point increases, we feel cold until our voluntary and involuntary actions cause our temperature to increase to this new level. We can decrease heat loss by covering ourselves with blankets (which is voluntary) and decrease blood flow to the skin (involuntary). We can increase our metabolic rate by shivering (involuntary). As noted in Chap. 6, anesthesia can affect the temperature regulation mechanism of the body [5].

Women tend to prefer higher room temperatures than do men, by as much as ∼3 °C, even though both men and women are thermally comfortable with the same average skin temperature, ≃33 °C. In air-conditioned offices temperatures are set on the basis of models that use the higher resting metabolic rates of men, and so women tend to feel cold in such offices [8, 13, 14, 20].

With advancing age, there is reduced metabolic production of heat and peripheral vasoconstriction in response to low environmental temperatures, and reduced sweat output and skin blood flow, and increased cardiac output in response to high environmental temperatures. However, the temperature regulatory system is not very comprised with advancing age, after accounting for the effects of chronic disease and the more sedentary lifestyle [11].

13.2.2 Control of Blood Pressure

The brain continuously receives input from the heart and muscles and then sends signals to control the heart and blood vessels to control blood pressure (Fig. 13.6). For example, the carotid sinus baroreceptors (pressure sensors in the carotid artery) fire a signal that is transmitted by the carotid sinus nerve that goes to the brain [18]. The frequency of this signal F is roughly proportional to how much the arterial pressure—as sensed by the baroreceptors p_b—exceeds a threshold pressure p_t, and this dependence has been modeled as

$$F + K_1 \frac{dF}{dt} = K_2(p_b - p_t) + K_3 \frac{dp_b}{dt}, \tag{13.34}$$

where K_1, K_2, and K_3 are constants. Note the similarity of this equation to that describing the Kelvin model of the mechanical properties of materials (4.68). The frequency is not directly proportional to the excess pressure because of the two derivative terms. The dp_b/dt term describes the observed overshoot in frequency.

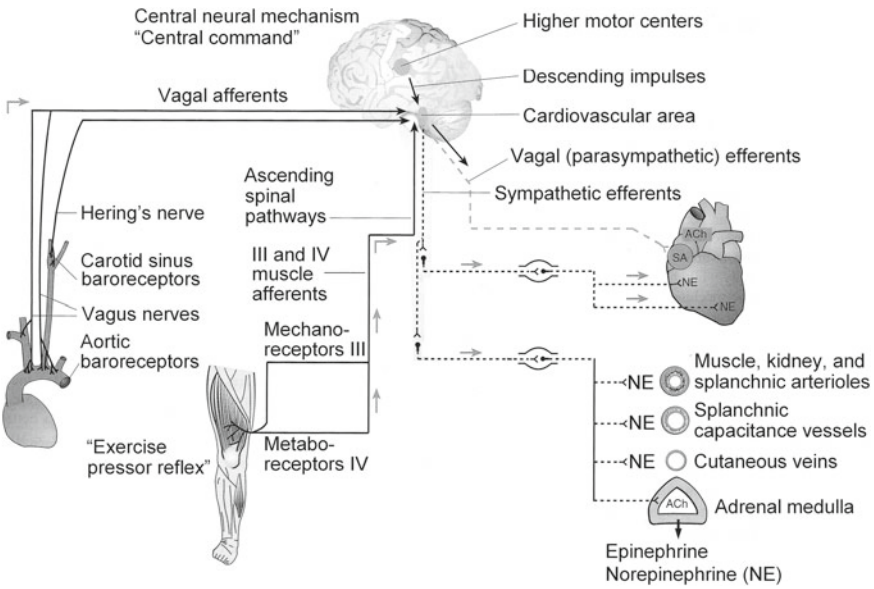


Fig. 13.6 Control of the cardiovascular system during exercise, showing control inputs on the *left* and outputs on the *right*. Nerve impulses from the motor region of the cerebrum in the brain, and the afferent input from the arterial baroreceptors and skeletal muscle receptors converge on the cardiovascular region of the medulla in the brain. During exercise this results in reduced parasympathetic activity to the heart and increased sympathetic activity to the heart, blood vessels, and the adrenal medulla. (The parasympathetic and sympathetic nervous systems are automatic regulation systems that, respectively, "relax" and "push" systems, such as organs, in the body.) The leads to an increase in blood pressure and cardiac output (From [4]. Used with permission)

The $\mathrm{d}F/\mathrm{d}t$ term describes how the rates of receptor firing continue to change even after the pressure has stabilized.

In the overall feedback system there is a decrease in arterial pressure Δp_a observed in response to a change in the pressure in the baroreceptors Δp_b, which is given by

$$\Delta p_\mathrm{a} = -g\Delta p_\mathrm{b}, \tag{13.35}$$

where g is the gain factor. A more exact expression is

$$\Delta p_\mathrm{a} = -g\,\Delta p_\mathrm{b} - K_4 \frac{\mathrm{d}p_\mathrm{b}^+}{\mathrm{d}t}, \tag{13.36}$$

where the last term denotes the rate of change of pressure only when it is increasing (p_b^+). The open loop gain factor g_o has been measured to range from 2 to 9 in dogs and humans in isolated systems with no feedback. The factor in a closed loop system with feedback g_c can be smaller; in particular the body makes it smaller during hemorrhaging.

13.2.3 Regulation During Exercise

Several functions and properties of the body are under active control during exercise. Arterial baroreceptors and skeletal muscle mechanoreceptors are inputs to the cardiovascular area of the medulla in the brain, along with impulses from the motor region of the cerebrum, to regulate the cardiovascular system (Fig. 13.6). This results in sympathetic activity in the heart, blood vessels, and adrenal medulla, which leads to increases in blood pressure and cardiac output.

For this increased blood flow to be useful, there must be a concomitant increase in air intake and therefore control of the ventilation system is vital (Fig. 13.1). In addition to the signals used for cardiovascular control are signals from receptors that control the respiratory components, such as the intercostal muscles and diaphragm. This results in neural activity to increase the rate and depth of breathing, which regulates the arterial O_2, CO_2, and pH. (See [9, 10] for a model of breathing.) Of course, during exercise the body must control temperature (Fig. 13.4).

During exercise the metabolism rate increases and the body must actively lose heat. This occurs through the temperature control system, outlined above and in Fig. 13.4.

13.2.4 Regulation of the Pupil Size

The size of the pupil of an eye and the pupil reflex are controlled by the contractions of the circularly-arranged pupillary constrictor muscle and the radially-arranged

pupillary dilator muscle [10]. This regulation is controlled by the light (luminous) flux through the pupil $\phi = IA$, where I is the illuminance on the eye and A is the area of the pupil. In steady state, the higher I is, the smaller A becomes, so this is an example of negative feedback. However this feedback occurs after a delay time or latency of $\tau \sim 180 - 400\,\text{ms}$, due to delay times in nerve transmission from the retina to the brain, analysis in the brain, nerve transmission from the brain to these muscles, and the activation of the muscles.

We can model this muscle activity $x(t)$ by relating it to the rate of arriving action potentials from the brain $R(t)$

$$\tau_x \frac{\mathrm{d}x(t)}{\mathrm{d}t} + x(t) = R(t), \tag{13.37}$$

where τ_x is this specific response time. It is known that $R(t)$ is zero below a given light flux threshold ϕ_{th} and increases logarithmically with flux above this threshold, for this flux a time τ earlier, $\phi(t-\tau) = I(t-\tau)A(t-\tau)$. So $R(t)$ is proportional to $F(u)$, with $F(u) = u$ for $u > 0$ and $= 0$ for $u \leq 0$, where $u = \ln(\phi(t-\tau)/\phi_{\text{th}})$; this form accounts for there being no response below the flux threshold.

The decreasing size of the pupil area A with muscle activity can be modeled by

$$A(x) = A_0 + A_1 \frac{\theta^n}{x^n + \theta^n}, \tag{13.38}$$

where A_0 is the minimum iris area (that occurs for the largest muscle activity $x(t)$), $A_0 + A_1$ is the maximum pupil area (that occurs for the smallest $x(t)$), and θ and n are constants. This equation can be solved to give $x(A)$.

The overall feedback and control equation is then

$$\tau_x \frac{\mathrm{d}x(t)}{\mathrm{d}t} + x(t) = \gamma F\left(\ln \frac{I(t-\tau)A(x(t-\tau))}{\phi_{\text{th}}}\right), \tag{13.39}$$

where γ is a constant. Solving for $x(t)$ gives the iris area $A(t)$ by using the relation for $x(A)$.

For a constant illuminance $I(t) = I_0$ this equation has a steady-state solution given by

$$x = \gamma F\left(\ln \frac{I_0 A(x)}{\phi_{\text{th}}}\right). \tag{13.40}$$

By using an analysis very similar to the control theory earlier this chapter, we can show that this solution is stable for a range of I and τ and unstable outside this range [10]. For large values of I or τ the iris area oscillates sinusoidally. This behavior can be observed by shining a focused light (but please not a laser!) at the edges of the pupil. The pupil contracts in response to the high light level thereby blocking this light, but then the pupil expands because little light is passing through it, and the oscillation continues.

13.3 Summary

Feedback and control are essential to every aspect of normal body function and regulation, and these processes can be modeled. Models of controlling body temperature by passive and active processes can be developed by using the models of heat formation and loss presented in Chap. 6.

Problems

13.1 Use the decision-making boxes as in Fig. 13.2 to devise a feedback and control system that redirects the flow of blood superficial or deep in the body for temperature regulation, as in the countercurrent mechanism described in Chap. 8.

13.2 Someone has a body temperature set point of 37.0 °C. Her temperature feedback and control system turns on whenever her temperature strays by ±0.5 °C from this set point, and returns it to the set point in 2 min. Unfortunately, when this feedback system is not on, her temperature drifts by +2 °C/h. How often does her feedback system turn on?

13.3 What is the maximum gain for stable feedback and control if the delay is alternatively 1 s or 1 ms?

13.4 (a) Your 70 kg body stops generating heat (but somehow functions well anyway), while you are indoors under normal conditions. Estimate how long it will take for you to cool from 37 to 28 °C (and die).
(b) As your temperature approaches 28 °C in (a) your metabolism turns on again and your metabolic rate is your normal BMR (Chap. 6). Estimate how long it will take for your body temperature to reach 37 °C again.
(c) Repeat (b) if your metabolic rate is at its summit value (Chap. 6).

13.5 Solve (13.33) for an arbitrary activity level.

13.6 Explain why the inhibition of shivering and vasoconstriction (the ability to constrict blood vessels) and the reduced metabolic rate that occur during general anesthesia, hurt the ability of the body to control temperature [5].

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